

Gursneh Kaur

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

JK Lakshmipat University, Jaipur — B.Tech, Computer Science and Engineering

Aug 2023 – May 2027

CGPA: 8.74/10.0 | Visiting Student, IIT Gandhinagar — Semester IV Exchange (2025)

Coursework: DSA, Operating Systems, Computer Networks, TOC, DBMS, Machine Learning, Deep Learning

Technical Skills

- **Languages:** C, C++, Python, JavaScript, Rust, x86 Assembly (NASM)
- **Systems & OS:** Preemptive scheduling, memory management, syscall interfaces, ELF loading, filesystems, GRUB/multiboot, QEMU
- **ML / CV / AI:** TensorFlow, PyTorch, OpenCV, YOLO, MTCNN, NLP, computer vision pipelines
- **Web & Mobile:** React.js, React Native (Expo Router), Node.js, Express.js, REST APIs, JWT/Authentication
- **Databases:** MySQL, MongoDB, PostgreSQL, Redis, SQLite
- **Cloud, DevOps & Tools:** Docker, Git, Linux, Wireshark, Salesforce (Apex, Visualforce)

Experience

AI Intern — AICTE Edunet Foundation [Certificate]

Oct 2025 – Nov 2025

- Built a real-time computer vision system for adaptive traffic signal control (YOLO-based vehicle detection), achieving a 17.8% improvement in traffic flow efficiency with adaptive green time of 15–27s within a 90–95s cycle.
- Engineered a vehicle counting and density-analysis pipeline processing 15+ FPS in real time, computing optimal green-light duration from live queue length and flow patterns.

Web Developer — AI ShipShape Pvt. Ltd. (UK) [Certificate]

May 2025 – Jul 2025

- Built and deployed 3+ full-stack web applications (React.js, Node.js, Express.js, MongoDB) with a RESTful API architecture using JWT authentication and role-based access control, serving real production users.

Salesforce Developer Intern — MTX Group [Certificate]

May 2025 – Jul 2025

- Automated enterprise business workflows using Apex and Visualforce and built custom CRM dashboards/reports, reducing manual effort and improving data visibility for client teams.

AI Vicharana Shala Bootcamp Research Intern — IIT Ropar [Certificate]

May 2024 – Jul 2024

- Mastered foundational AI concepts under Prof. Sudarshan Iyengar (HOD, CSE) through mathematical intuition and theoretical frameworks spanning statistical learning and optimization theory.
- Implemented machine learning algorithms and neural networks from scratch, achieving 95% accuracy on benchmark datasets.

Projects

myOS — x86 Operating System Built From Scratch [GitHub]

2026

- Designed and implemented a hobby x86 OS in C and NASM assembly, booting via GRUB multiboot and running on QEMU.
- Built a preemptive scheduler with context switching, Ring 3 user-mode execution, and a full syscall interface bridging user and kernel space.
- Implemented an ELF loader, a ramdisk filesystem, and a paging-based heap allocator; built an interactive shell supporting process execution.

Visual Scene Explainer for Blind Persons [GitHub]

2025 – 2026

- Developing an accessibility-focused AI tool combining real-time computer vision and NLG to generate contextual, human-readable audio scene descriptions from live camera frames.

Green Light Traffic Analyzer — Extended System [GitHub]

2025 – 2026

- Extended the AICTE traffic-signal system into a production-grade platform: replaced fixed-cycle logic with a pressure-score model inspired by SCOOT/SCATS.
- Added emission modeling, a live processing pipeline, an interactive observability dashboard, and config/logging infrastructure; validated with a 42-test pytest suite.

Biometric Face Recognition System [GitHub]

2025

- Built real-time face recognition using MTCNN and a deep learning model generating 128-dimensional face embeddings; identity verified via Euclidean distance on live webcam feed.

Other Projects

- **LenDen [GitHub]** — cross-platform community marketplace app with Razorpay payments and role-based access (React Native, Node.js/Express.js, MongoDB)
- **Event Vibes [GitHub]** — full-stack event media platform serving 200+ students across 50+ annual events (React.js, Node.js, MongoDB, Cloudinary)

Achievements & Certifications

- Google Build With India Challenge (2025) — Ranked top 5,000 of 25,000+ repositories for a full-stack platform centralizing faculty tasks.
- Merit Scholarship, JKLU — Awarded 50%; increased to 100% through consistent academic performance (2023–2027).
- Teaching Assistant — Python and DAA courses, JKLU (Jul–Dec 2024).
- Accenture Data Analytics & Visualization — Forage virtual experience program (2024).